

DIVISION S-7—FOREST & RANGE SOILS

Carbon and Nitrogen Dynamics in Preferential Flow Paths and Matrix of a Forest Soil

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ABSTRACT

Natural abundance (δ) of the stable isotopes ^{13}C and ^{15}N has gained acceptance for studying C and N cycling in forests. In most studies, bulk soil samples are collected to determine isotope abundance. Such sampling overlooks the potential impact of preferential flow on isotope distribution. The objective of this study was to investigate the effects of preferential flow on the distribution of soil organic carbon (SOC), total N, $\delta^{13}\text{C}$, and $\delta^{15}\text{N}$ in a forest soil in Central Switzerland. Preferential flow paths in the soil were identified with a dye tracer, Brilliant Blue (Plüss-Stauffer AG, Oftringen, Switzerland), that was homogeneously applied to the soil surface. In the stained preferential flow paths, concentrations of SOC and total N were 15 to 75% higher than in the soil matrix. The total increase of SOC in preferential flow paths ranged from 740 to 960 g C m $^{-2}$ in four individual soil plots. Values of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were lowest in tree leaves and in the forest floor, and increased with soil depth, thus with the degree of decomposition of SOC. In the mineral soil, preferential flow paths were significantly depleted in ^{13}C by 0.15 to 0.4‰ as compared with the soil matrix. The $\delta^{15}\text{N}$ values increased with soil depth from 0.9 to 4.7‰ in the preferential flow paths and from 0.5 to 6‰ in the soil matrix. Adding a highly enriched ^{15}N -tracer homogeneously to the soil surface showed a higher recovery of ^{15}N in the soil and in the fine roots sampled from preferential flow paths than in those sampled from the soil matrix. Our results suggest that in preferential flow paths, SOC is younger and N cycling is more rapid than in the soil matrix.

THE STABLE ISOTOPES OF $\delta^{13}\text{C}$ AND $\delta^{15}\text{N}$ has been widely used to study C and N cycling in forest soils (Nadelhoffer and Fry, 1988; Balesdent et al., 1993; Högberg, 1997). The aim of many of these studies was to gain information about the transformation and turnover of soil organic matter (SOM) and to determine the age of SOM with help from ^{14}C dating or to assess forest ecosystem health (Gebauer and Schulze, 1991). Since the heavier isotopes are discriminated against during biological and chemical processes, they become enriched in the substrate compared with the product. Therefore, the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values generally increase with soil depth and degree of decomposition of SOM (Nadelhoffer and Fry, 1988; Balesdent et al., 1993; Huang et al., 1996; Bol et al., 1999).

In all of these studies, bulk samples are taken to determine the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in soils. However, the procedure of taking depth-wise bulk soil samples averages over a certain soil volume and disregards spatial hetero-

geneity. Heterogeneity is considered in the vertical direction by sampling different soil depths, and in the horizontal direction on larger scales by sampling different plots or compartments of ecosystems. On the intermediate scale of centimeters to meters, however, heterogeneities exist as well. These heterogeneities are not necessarily randomly distributed but are likely associated with soil structure, which is important for microbial processes and nutrient cycling (Parkin, 1993; Hagedorn et al., 1999).

Preferential flow is the rapid transport of water and solutes through the soil that bypasses a large portion of the soil volume (Beven and German, 1982; Flury and Flühler, 1994b). It comprises flow through macropores such as root channels, earthworm (*Lumbricus terrestris*) channels, mouse (*Microtus arvalis*) burrows, fissures, and cracks, as well as fingered flow due to wetting front instabilities (Stagnitti et al., 1995; Noguchi et al., 1999). In soils, preferential flow is the rule rather than the exception (Flury and Flühler, 1994b). In catchments, lateral preferential pathways are important determinants in subsurface solute transport (Tsuboyama et al., 1994; Montgomery et al., 1997; Uchida et al., 1999). Preferential flow is also relevant for transport of dissolved organic matter to deeper soil depths (Jardine et al., 1989; Hagedorn et al., 2000).

Thus, preferential flow paths may act as transport pathways of young SOM into the deeper soil. Furthermore, it was shown that preferential flow paths have an increased microbial biomass as compared with the soil matrix, since locations along flow paths are more exposed to drying and wetting, and have a better nutrient supply (Bundt et al., 2001). As a consequence, preferential flow paths might be locations with an enhanced turnover of SOM and nutrients.

In this study, we investigated the SOC, total N, and the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in preferential flow paths and in the soil matrix and studied the fate of ^{15}N -double labeled NH_4NO_3 in both soil compartments. Our objective was to test the hypothesis that the isotopic signatures of preferential flow paths differ from the rest of the soil, which would point to different SOC- and N-turnover rates and to different C and N input in these flow regions.

MATERIALS AND METHODS

Study Site

The study was conducted on four plots situated approximately on the corners of a 2-ha site in a spruce [*Picea abies* (L.)

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Abbreviations: DOC, dissolved organic carbon; SOC, soil organic carbon; SOM, soil organic matter; C_{org}, organic carbon; δ , natural abundance.

Karst.]-beech (*Fagus sylvatica* L.) forest in Unterehrendingen, Central Switzerland (47°30'34"N lat., 008°20'50"E long.). The topography was flat and had a slope of <2%. The forest was planted in 1930 and was managed by selective thinning. The soil type was a Typic Haplumbrept (Soil Survey Staff, 1994) and developed on Upper Marine Molasse overlain with end-moraine at one of the four plots. The soil texture was sandy loam, and the soil was well-drained, although from 50 cm down, it showed signs of redoximorphosis. From 1992 to 1999 the mean annual precipitation was 1120 mm.

Identification of Preferential Flow Paths and Sampling

We identified preferential flow paths in the soil by applying the food dye Brilliant Blue FCF (CI. 42090) with a field sprinkler. The amount of 45-mm dye solution (deionized water containing 3 g L⁻¹ Brilliant Blue) was homogeneously distributed to the soil surface of a 1 by 1.5 m plot during 6 h (Flury and Flühler, 1994a,b) (Fig. 1). One day after dye application, we opened a trench of 1.2-m depth. A vertical soil profile of 1 by 1 m was prepared 0.3 m away from the plot's border within the sprinkled area. Photographs were taken that were later used to estimate the dye coverage, corresponding to the volumetric proportion of preferential flow paths. Blue-stained areas were defined as preferential flow paths, nonstained areas as matrix. The sampling depths 0 to 9, 9 to 20, 20 to 50, and 50 to 100 cm were defined approximately according to morphological soil horizons. In each depth, we took soil samples over the whole horizon with a small spatula from the stained preferential flow paths and from the unstained soil matrix. Root samples were collected from six small cores (vol. = 9.1 cm³) taken within each depth on each plot. Three cores were pushed by hand into the blue-stained preferential flow paths, three into the unstained soil matrix. After recovery, the cores were stored at 4°C until further treatment.

Four more profiles were prepared at 10-cm distance from the previous profile and the sampling procedure was repeated. The samples of the five profiles were later pooled to obtain one composite sample per depth and flow region for each of the four soil plots. On each plot, the organic layer was sampled completely and then separated into Oi, Oe, and Oa layers. To estimate the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, samples were taken from all four plots in April 1998.

On three of the four plots, three ceramic suction cups were installed at 15-, 25-, and 80-cm depths (nine cups total per plot) in December 1997. The suction cups were allowed to

equilibrate with the soil for 4 mo. Following this equilibration period, soil solution samples were collected every 2 wk from May to December 1998 using a small self-regulating motor that maintained a constant suction of 600 kPa. Additionally, three Plexiglas zero-tension lysimeters were installed in each plot to sample the free draining water percolating through the organic soil horizon. Samples collected during this May to December period from both the tension and zero-tension lysimeters were pooled on a volume proportional basis for natural isotope abundance analysis.

Nitrogen-15 Labeling Experiment

To study the fate of deposited N in the soil, a ¹⁵N-tracer experiment was conducted on all four soil plots. In May 1998, we applied 20.15 mg ¹⁵N m⁻², which is equivalent to 55 mg double labeled and 99.8% enriched ¹⁵NH₄¹⁵NO₃. The tracer was dissolved in deionized water and distributed evenly with a vaporizer over the designated soil plots covering an area of 24 m² in total. The samples were taken as previously described in June and October 1998 and in May 1999.

Sample Preparation and Analyses

Soil and organic layer samples were oven-dried (50°C), sieved (2 mm), and ground to fine powder prior to analysis. Roots were washed out of the fresh soil, dried and also ground to fine powder. Root samples of all four soil plots were pooled to obtain enough material for the analysis. Soil solution samples were freeze-dried.

Total C and N of soil, organic layer, and root samples were determined with a CN auto analyzer (NA 1500, Carlo Erba Instruments, Italy). Dissolved organic carbon (DOC) was measured with a TOC-500 analyzer (Shimadzu Corp., Kyoto, Japan).

Potentially existing carbonates were removed prior to stable isotope analysis. An amount of 5 g per sample was suspended in 15 ml of distilled water and 25 mL of 1M HCl were added. After 30 min of stirring, when the pH reached 1, we added 1M KOH until the pH was between 4 and 5. After vacuum filtration, the sample on the filter paper was dried for 24 h at 65°C and then crushed to a fine powder. For the stable isotope analysis (¹³C and ¹⁵N), the ground soil, organic layer, soil solution, and root samples were weighed into small tin capsules. The sample amount for soil varied between 30 to 80 mg for N, 12 to 15 mg for C, and ~5 mg for root and organic layer C and N analyses. For the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ determination, the samples were combusted in an Elemental Analyzer (EA-1110, Carlo Erba, Italy) which was connected to a continuous flow mass spectrometer (DELTA-S, Finnigan MAT, Germany). The isotopic signatures are expressed in the delta notation:

$$\delta x = 1000 \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right) \quad [1]$$

with δx being the isotope ratio of C or N in delta-units relative to the international standards (Pee Dee Belemnite for C and atmospheric N₂ for N) and R_{sample} and R_{standard} are the ¹³C/¹²C or ¹⁵N/¹⁴N ratios of the samples and standards, respectively. The internal precision was for $\delta^{13}\text{C} \leq 0.1\%$ and for $\delta^{15}\text{N} \leq 0.15\%$.

Data Analysis

Concentrations of SOC and total N in the preferential flow paths were corrected for the C and N added with the experimental Brilliant Blue application. The dye-tracer Brilliant Blue had a C content of 56%, and a N content of 3.3%. The

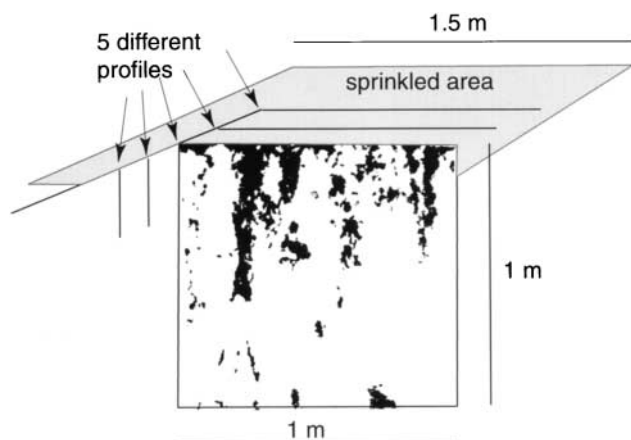


Fig. 1. Experimental setup and processed photo of one of the soil profiles with preferential flow paths being visualized by the dye-tracer Brilliant Blue.

total amount of C and N added with Brilliant Blue was 75 g C m⁻² and 4.5 g N m⁻², respectively. The Brilliant Blue content of the soil was estimated by measuring the Brilliant Blue concentrations in soil extracts with 0.5 M K₂SO₄ and assuming a mass recovery of 0.2, based on the lowest recovery reported in other studies (Forrer, 1998). Since the estimate of the total Brilliant Blue input by this procedure is about 50% larger than the amount of applied Brilliant Blue, this is a conservative estimate (i.e., we use the worst case estimates to be sure that we do not underestimate the amount of Brilliant Blue derived C).

The potential change in ¹³C and ¹⁵N of the soil due to the applied Brilliant Blue was calculated as follows:

$$\Delta\delta = f_{\text{BB}}(\delta_{\text{BB}} - \delta_{\text{soil}}) \quad [2]$$

where $\Delta\delta$ is the potential change of the soil $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ through the Brilliant Blue addition, f_{BB} is the maximal fraction of Brilliant Blue derived C and N at SOC and at total N, δ_{BB} and δ_{soil} represent the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of Brilliant Blue and the soil, respectively. Brilliant Blue had a $\delta^{13}\text{C}$ value of -27.5‰ and a $\delta^{15}\text{N}$ value of -1‰.

The enrichment of SOC in preferential flow paths as compared with the soil matrix (SOC_e) was calculated as follows:

$$\text{SOC}_e = f_i V \rho_s (C_i - C_m), \quad [3]$$

where f_i is the fraction of identified preferential flow paths of the total soil volume, V ; ρ_s is the soil bulk density, and C_i and C_m are the SOC concentrations in the preferential flow paths and in the soil matrix, respectively. The SOC concentrations of the preferential flow paths were corrected for their Brilliant Blue content as described above. The enrichment of fine root C along preferential flow paths was calculated with the same equation using the concentrations of fine root C in the soil. To estimate the potential contribution of DOC to the C enrichment in the preferential flow paths, we estimated the drainage with a simple water budget, where drainage equals the difference between rainfall and evapotranspiration. Evapotranspiration was ~550 mm according to Spreafico et al. (1992).

The differences of parameters between preferential flow paths and soil matrix were tested for statistical significance with analysis of variance or for single depth zones with the paired *t*-test. Residuals were tested for normality and for similar variance. In case of nonnormality, we preferred the more robust nonparametric Wilcoxon signed-rank test. All calculations were conducted with the software S-plus (version 5.0, MathSoft, Seattle, WA).

RESULTS

Flow Patterns

An example of the heterogeneous dye distribution as the result of heterogeneous water flow is given in Fig. 1. The proportions of stained preferential flow paths at the total soil volume ranged from 70 ± 11% in 0- to 9-cm depth to 2 ± 2% at 50- to 100-cm depth. Our experimental approach did not distinguish between different causes for preferential flow like macropore flow, heterogeneous infiltration, wetting front instabilities or flow along roots. However, we noticed that most of the preferential flow paths coincided with living or decayed roots. The other major cause of preferential flow in the studied forest soil was flow along structural voids like cracks or fissures. Because of the low soil pH, there were no wormholes or other animal burrows.

Table 1. Physical and chemical properties of preferential flow paths and soil matrix.

Depth	Flow region	pH	Sand	Silt	Clay	N _{tot}	C _{org}	C/N	Root density
cm			— % —			— g kg ⁻¹ —			mg L ⁻¹
0–9	Preferential	3.4	48	34	18	2.18**	34.5**	15.83	1654**
9–20	flow	3.7	47	34	19	0.98*	14.5**	14.80	859*
20–50	path	3.8	44	38	18	0.65*	8.2**	12.61	645**
50–100		3.8	28	45	27	0.47*	4.9**	10.42*	n.d.
0–9	Matrix	3.4	41	38	21	1.57	24.3	15.48	1022
9–20		3.6	40	39	21	0.88	12.9	14.66	508
20–50		3.8	40	40	20	0.60	6.4	10.67	226
50–100		3.8	30	44	26	0.34	2.9	8.93	n.d.†

* Significance at the 0.05 probability level.

** Significance at the 0.01 probability level.

† Not determined.

Soil Organic Carbon

The organic C (C_{org}) content of the soil showed a sharp decrease with depth ($P < 0.001$), and was higher in the preferential flow paths than in the matrix at all depths (Table 1). Soil organic C was increased in preferential flow paths compared with the soil matrix by 42% at 0- to 9-cm and 12, 28, and 69% in the subsequent depths. The experimental addition of Brilliant Blue increased the C_{org} content of the preferential flow paths maximal by 2.5% of SOC in 0- to 9-cm depth and by maximal 3.5% of SOC in 50- to 100-cm depth. However, for the calculations of the total SOC enrichment, we subtracted the Brilliant Blue derived C from the total C_{org} concentration. Summing up over the total soil compartment, and taking the volume of preferential flow paths into account, yields a total SOC accumulation along preferential flow paths of 740 to 960 g C m⁻² in the four plots with a mean accumulation of 860 g C m⁻².

The needles of the dominant tree species in the forest, Norway Spruce were markedly depleted in ¹³C compared with the mineral soil. The mean $\delta^{13}\text{C}$ value of needles was -27.0‰ (needles up to 4 yr of age at time of sampling in 1998), single values of fine roots ranged between -26.8 and -28.9‰ (for mean values see Fig. 2). Fine roots sampled from the matrix were depleted in ¹³C as compared with the fine roots from the preferential flow paths ($P < 0.001$). Most depleted in ¹³C was the forest floor with $\delta^{13}\text{C}$ values of -28.0 to -28.5‰.

In the mineral soil, the $\delta^{13}\text{C}$ values increased significantly with depth ($P < 0.001$) from -26.8 to -25.8‰ in the preferential flow paths and from -26.5 to -25.4‰ in the matrix (Fig. 2). Across all depths, SOC of the preferential flow paths was significantly more depleted in ¹³C than that of the matrix soil ($P = 0.001$). This was not caused by the experimental addition of Brilliant Blue, because the fraction of Brilliant Blue derived C was negligible in comparison with the SOC content of the soil (~2.5% of SOC in 0- to 9-cm depth to 3.5% of SOC in 50- to 100-cm depth). The estimated maximal decrease in $\delta^{13}\text{C}$ according to Eq. [2] due to Brilliant Blue was below 0.05‰.

To estimate the accumulation of SOC with ¹³C in the soil profile, an isotopic discrimination factor (i.e., the slope of the regression between $\delta^{13}\text{C}$ and -ln [%C]) was calculated. The isotopic discrimination factor was originally derived for mineralization studies and shows

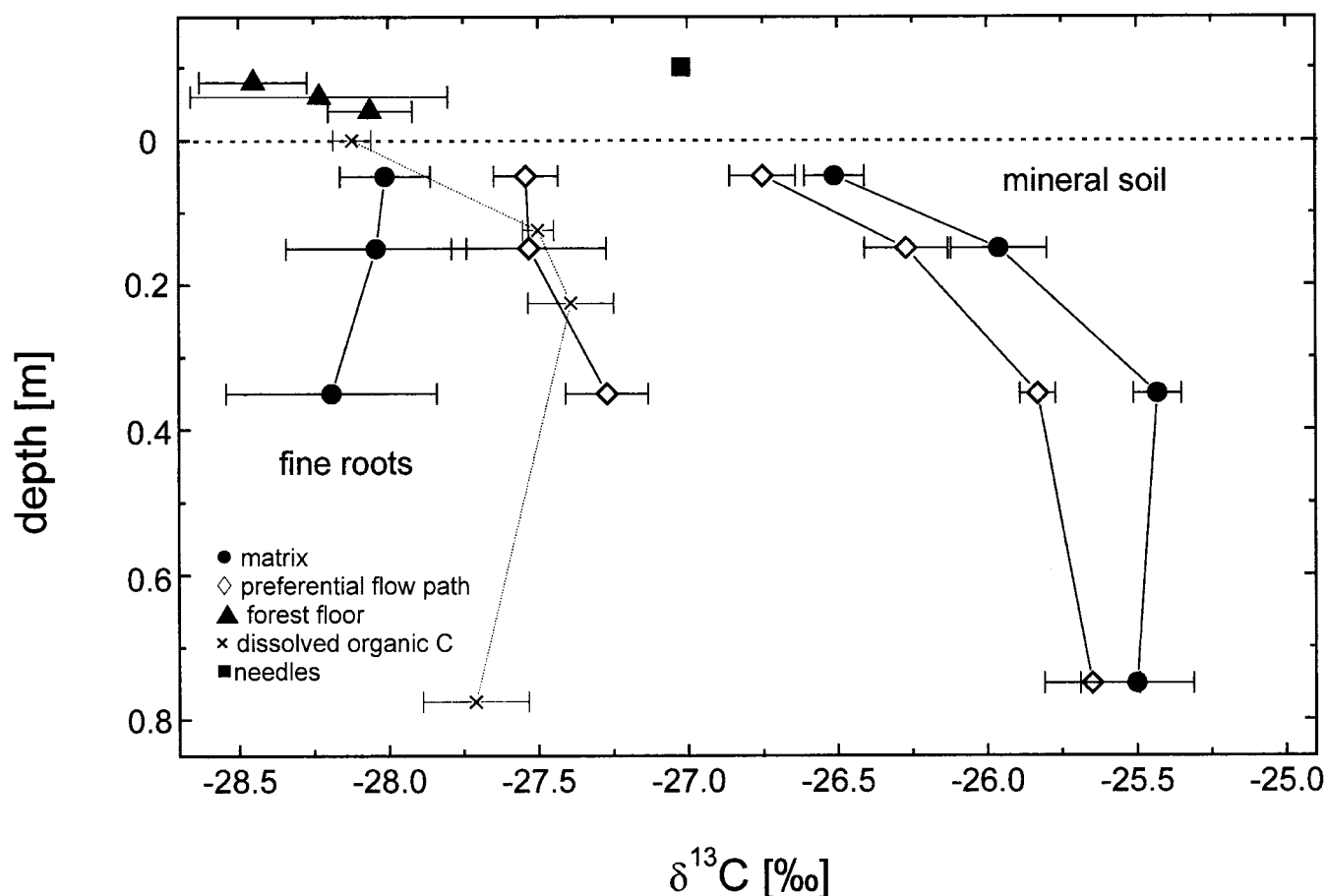


Fig. 2. The $\delta^{13}\text{C}$ values of the dominant tree species (Norway spruce), forest floor, and of mineral soil samples from the preferential flow paths and matrix, fine roots of these flow regions, and of the soil solution. Root isotopic signatures are the mean of five sampling dates. Standard errors are given as error bars.

the degree of isotopic discrimination during organic matter decomposition (O'Leary, 1981). It can also be applied to soil profiles, where the SOM content usually decreases with depth (Nadelhoffer and Fry, 1988; Balesdent et al., 1993). The discrimination factor was similar for preferential flow path and soil matrix samples (Fig. 3). However, the discrimination ratio, i.e., the ratio between $\delta^{13}\text{C}$ and $-\ln(\%C)$ of the individual samples, was significantly lower for preferential flow paths than for soil matrix (Wilcoxon signed rank test: $P = 0.004$).

In the soil solution, the concentrations of DOC decreased from 35 to 45 mg DOC L⁻¹ directly underneath the organic layer to <2 mg DOC L⁻¹ at an 80-cm depth (S. Zimmermann, WSL, personal communication). Down to 50-cm depth, the $\delta^{13}\text{C}$ signature of DOC showed about the same increase with depths as the mineral soil. However, the $\delta^{13}\text{C}$ values of DOC were about 2‰ more negative than those of SOC at all depths (Fig. 2).

Soil Nitrogen

Similar to the C contents, the total N contents in the mineral soil decreased significantly with sampling depths ($P < 0.001$, Table 1). Across all depths, the preferential flow paths had significantly higher total N concentrations than the soil matrix ($P = 0.009$). Fine roots from the matrix had higher N contents than fine

roots from the preferential flow paths, however, this difference was statistically not significant.

The $\delta^{15}\text{N}$ followed the pattern of ^{13}C (Fig. 4). Values of $\delta^{15}\text{N}$ were lowest in needles with -5.7‰ and ranged from -5.4 to -5.0‰ in the forest floor. Fine roots had $\delta^{15}\text{N}$ values almost as low as in the forest floor, with the lowest values at 20- to 50-cm depth. In 0- to 9-cm depth, the fine roots from the matrix were depleted in ^{15}N as compared with those from the preferential flow paths, but this difference was statistically not significant.

In the mineral soil, the $\delta^{15}\text{N}$ values increased with depth from -0.9 to 4.7‰ in the preferential flow paths and from 0.5 to 6‰ in the matrix (Fig. 4). Across all depths, the depletion of ^{15}N in preferential flow paths compared with the matrix was significant ($P < 0.001$). As with $\delta^{13}\text{C}$, the $\delta^{15}\text{N}$ was not significantly affected by the experimental Brilliant Blue addition. The estimated maximal shift in δ values according to Eq. [2] was $<0.04\text{‰}$ in the topsoil and $<0.25\text{‰}$ in the lowest depth. This is substantially lower than the measured difference between preferential flow paths and soil matrix.

The $\delta^{15}\text{N}$ of the soil solution did not follow the signature of the mineral soil. In the topsoil, the N in the soil solution had similar $\delta^{15}\text{N}$ values as the mineral soil, but was distinctly lower in the deeper soil horizons (Fig. 4). Assuming a DOC to dissolved organic nitrogen (DON)

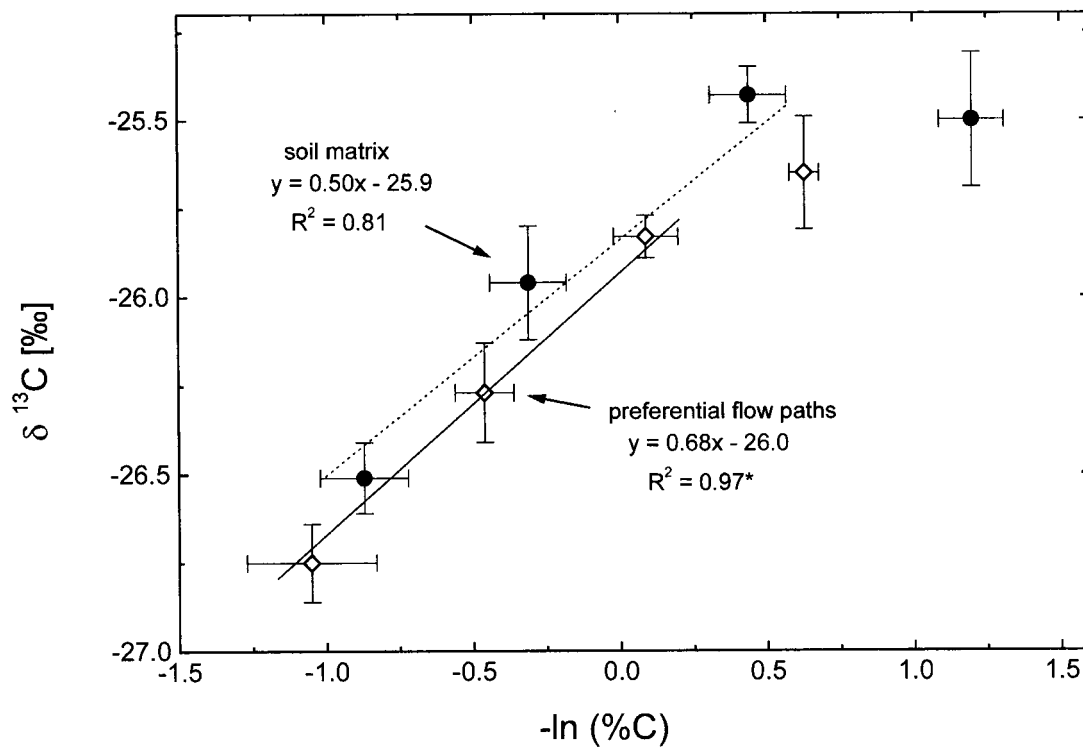


Fig. 3. Relationship between $\delta^{13}\text{C}$ values and soil organic carbon (SOC) concentrations in soil from the preferential flow paths and from the matrix. Means and standard errors of four plots. The slope of the respective regression line is called isotopic discrimination factor.

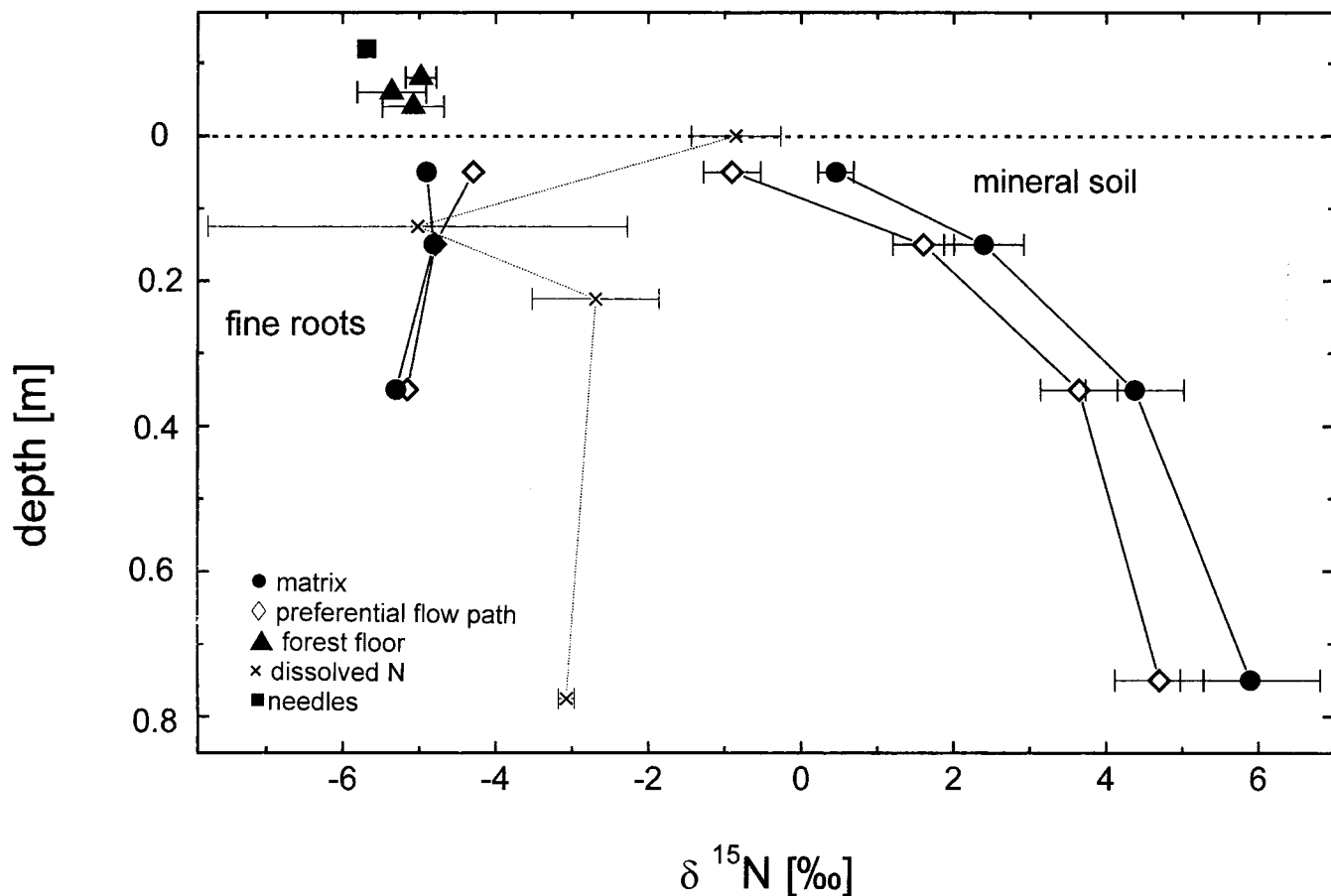


Fig. 4. The $\delta^{15}\text{N}$ values of the dominant tree species (Norway spruce), forest floor, and of mineral soil samples from the preferential flow paths and matrix, fine roots of these flow regions, and of the soil solution. Root isotopic signatures are the mean of five sampling dates. Standard errors are given as error bars.

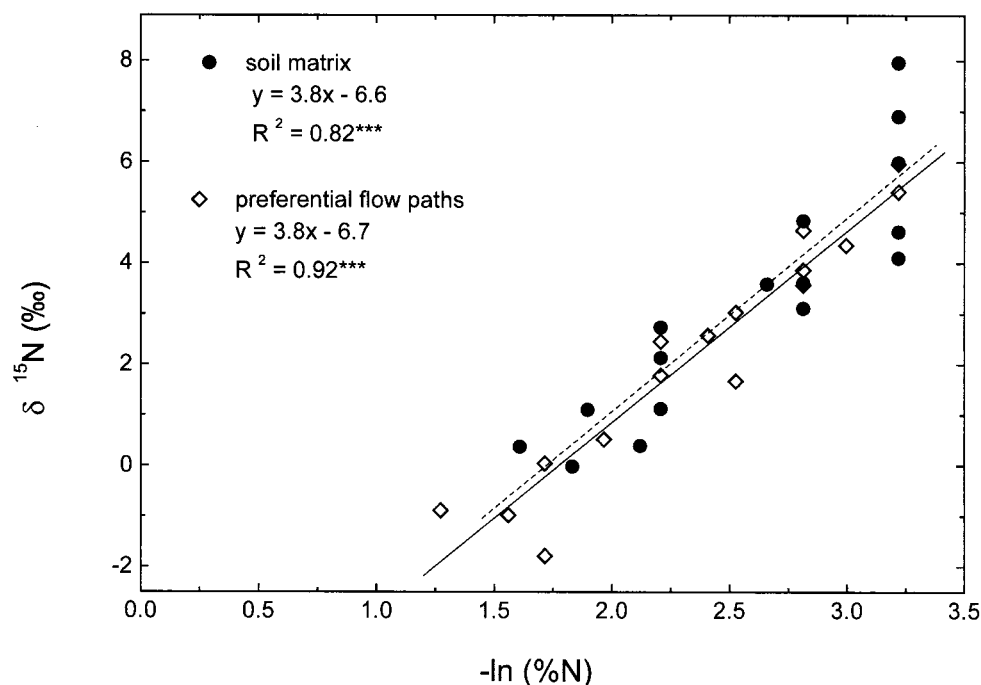


Fig. 5. Relationship between $\delta^{15}\text{N}$ values and total nitrogen concentrations in soil from the preferential flow paths and from the matrix. Means and standard errors of four plots. The slope of the respective regression line is called isotopic discrimination factor.

mass ratio of 30 (Currie et al., 1996; Hagedorn et al., 2000), the proportion of nitrate relative to the total dissolved N increased from 75% at the mineral soil surface to 100% in the soil solution below 25-cm depth. The discrimination factor for $\delta^{15}\text{N}$ was nearly identical for preferential flow paths and matrix (Fig. 5). However, the two regression lines were displaced horizontally, which is in accordance with the fact that the single discrimination ratios for preferential flow paths and matrix differed significantly (Wilcoxon signed rank test, $P < 0.001$).

Nitrogen-15 Tracer Study

The mean ^{15}N recovery of the ^{15}N -tracer experiment was $88 \pm 24\%$ after 1 yr, including roots, mineral soil, and forest floor. Most of the added ^{15}N was retained in the forest floor (60%). In the mineral soil, $\delta^{15}\text{N}$ values rose steeply after the tracer application (Fig. 6). The rate of increase in $\delta^{15}\text{N}$ was higher in the preferential flow paths than in the matrix (across all depths, $P = 0.01$). Subsequently, 6 and 12 mo after the ^{15}N addition, the $\delta^{15}\text{N}$ remained stable in the preferential flow paths at all depths, but still increased in the soil matrix of 9- to 20-cm and 20- to 50-cm depth.

A similar pattern was found for the $\delta^{15}\text{N}$ of fine roots (Fig. 6). The amplitude of the changes was about one order of magnitude higher in the fine roots than in the mineral soil. The temporal changes in $\delta^{15}\text{N}$ of the fine roots and of the soil were linearly correlated ($R^2 = 0.95$).

DISCUSSION

Soil Organic Carbon Dynamics

The increase of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ with soil depth is consistent with the results from other studies in forest soils

(O'Brien and Stout, 1978; Nadelhoffer and Fry, 1988; Balesdent et al., 1993; Koopmans et al., 1997). This increase is usually related to (i) overall discrimination against ^{15}N and ^{13}C during organic matter decomposition, (ii) differential preservation of SOM or plant litter components, (iii) changes from litter inputs with high $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values to litter inputs with lower values, and (iv) illuviation of ^{15}N and ^{13}C enriched dissolved organic matter into lower soil layers (Nadelhoffer and Fry, 1988). Our findings show that within a given horizon, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values were not evenly distributed. Preferential flow paths were depleted in ^{13}C and ^{15}N compared with the soil matrix.

Huang et al. (1996) and Bol et al. (1999) found that $\delta^{13}\text{C}$ values of SOC were closely correlated with ^{14}C values, indicating that $\delta^{13}\text{C}$ values increase with the age of SOC. Moreover, it was shown that labile C pools with high turnover rates (sand-associated and light SOC fractions) are depleted in ^{13}C compared with recalcitrant C pools (Balesdent et al., 1988; Bonde et al., 1992). Thus, our finding, that SOC of preferential flow paths was depleted in ^{13}C relative to the soil matrix strongly suggests that SOC of the preferential flow paths is younger than SOC of the soil matrix. This is supported by the relationship between $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values (Fig. 7). All potential input sources of SOC (leaves, forest floor, roots, and soil solution) fell into a relatively narrow range of about 2‰ in the direction of both x-axis and y-axis. In the mineral soil, the δ -values of C and N formed a straight line originating in the middle of the plant-input cloud (i.e., sources) and extending towards higher $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values. The uppermost horizon was closest to the plant-input source of the SOC. Within a given horizon, the values of the preferential flow paths always resembled the source more strongly than those

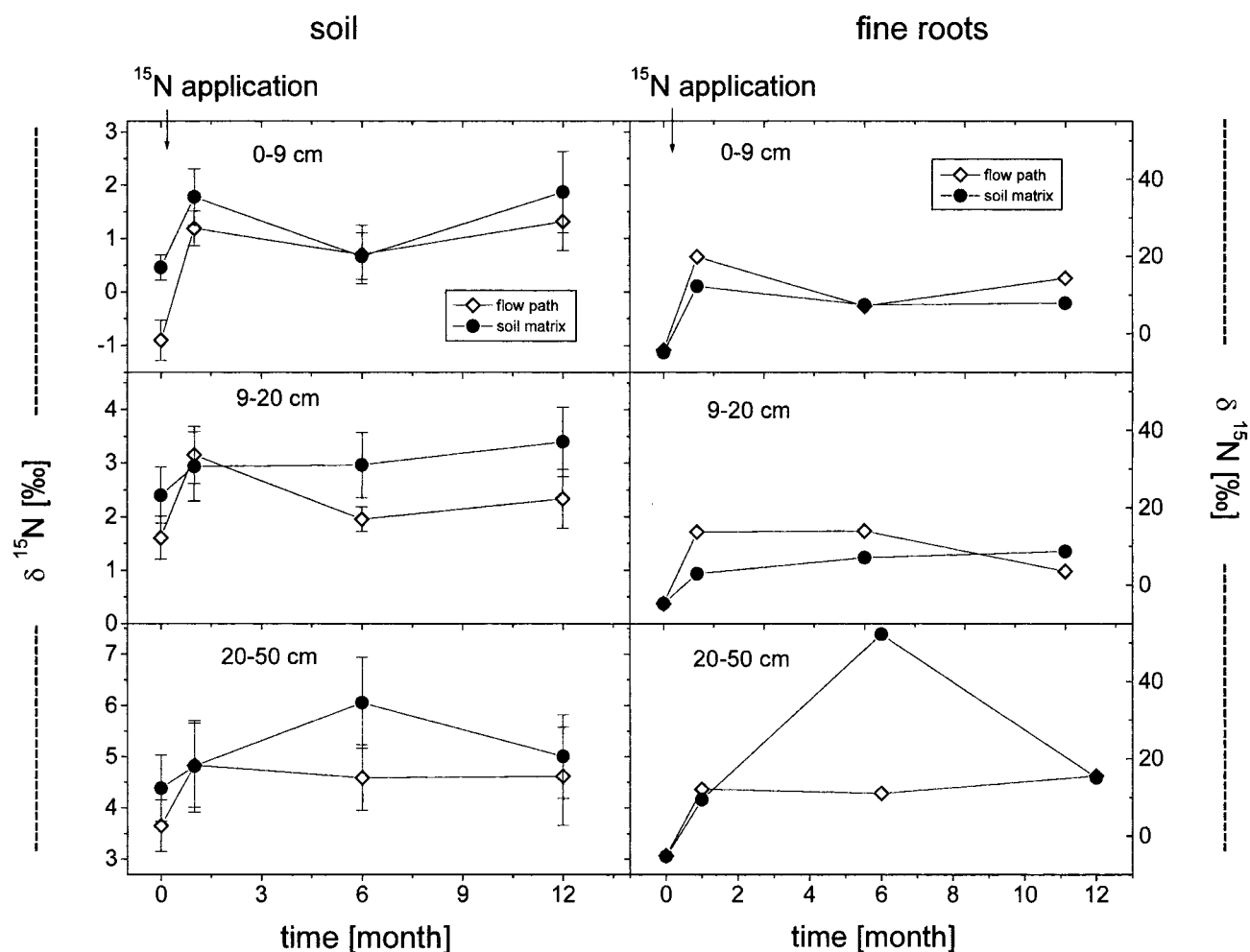


Fig. 6. Progression of $\delta^{15}\text{N}$ values in soil and fine roots of preferential flow paths and matrix before 1, 6, and 12 mo after ^{15}N application in 0- to 9-, 9- to 20-, and 20- to 50-cm depth. For the mineral soil, means and standard errors of three plots for the $\delta^{15}\text{N}$ of the soil are given. Root material from all four plots were pooled to yield enough material for analysis.

of the corresponding matrix soil. This suggests that SOC of preferential flow paths is younger and less humified.

The younger SOC in the preferential flow paths might be partly because of a higher input of root-derived C along preferential flow paths as indicated by a higher root biomass with rather negative $\delta^{13}\text{C}$ values (Table 1, Fig. 2). The C accumulation because of fine roots along preferential flow paths ranged between 20 and 40 g C m⁻² in the individual soil plots (estimated according to Eq. [3]). However, little is known about the relative amount of humus formed from roots or litter. Assuming a humification factor for fine roots of 0.5, which is about twice as high as those used in SOC models (Aber et al., 1990; Perruchoud et al., 1999) and a constant input per year, yields an annual contribution of the increased fine root biomass in the preferential flow paths of about 10 to 20 g C m⁻² yr⁻¹. This conservative estimate is almost two orders of magnitude lower than the measured SOC accumulation along preferential flow paths of 740 to 960 g C m⁻².

Increased illuviation of DOC and organic colloids might add to the inputs of young C even down to the deeper soil depths, because preferential flow paths are

preferred pathways for the transport of solutes and particles (Jardine et al., 1989; Hagedorn et al., 2000). The $\delta^{13}\text{C}$ values of DOC in the soil solution were close to those of fine roots and between 1 to 2‰ more negative than the mineral soil signature (Fig. 2). Transport of young C with percolating water was already shown for two Alfisols in Northern Germany by Becker-Heide-mann and Scharpenseel (1986) and is thought to be an important mechanism in pedogenesis (Buurman, 1985; Schoenau and Bettany, 1987; Blaser, 1994). Because DOC concentrations in the soil solution at 80-cm depth were <2 mg L⁻¹, we assume that almost all DOC that was leached from the forest floor was retained in the mineral soil. The maximal contribution of DOC leaching to the SOC accumulation in preferential flow paths ranged between 15 to 40 g C m⁻² yr⁻¹. This estimate is based on the following conservative assumptions: (i) that all of the drainage water (400–900 mm yr⁻¹ according to a simple water budget) moved through the identified preferential flow paths, (ii) that all of the DOC was sorbed to the walls of the flow paths, and (iii) that redistribution of DOC and mineralization of DOC was negligible. The estimated DOC leaching are consistent

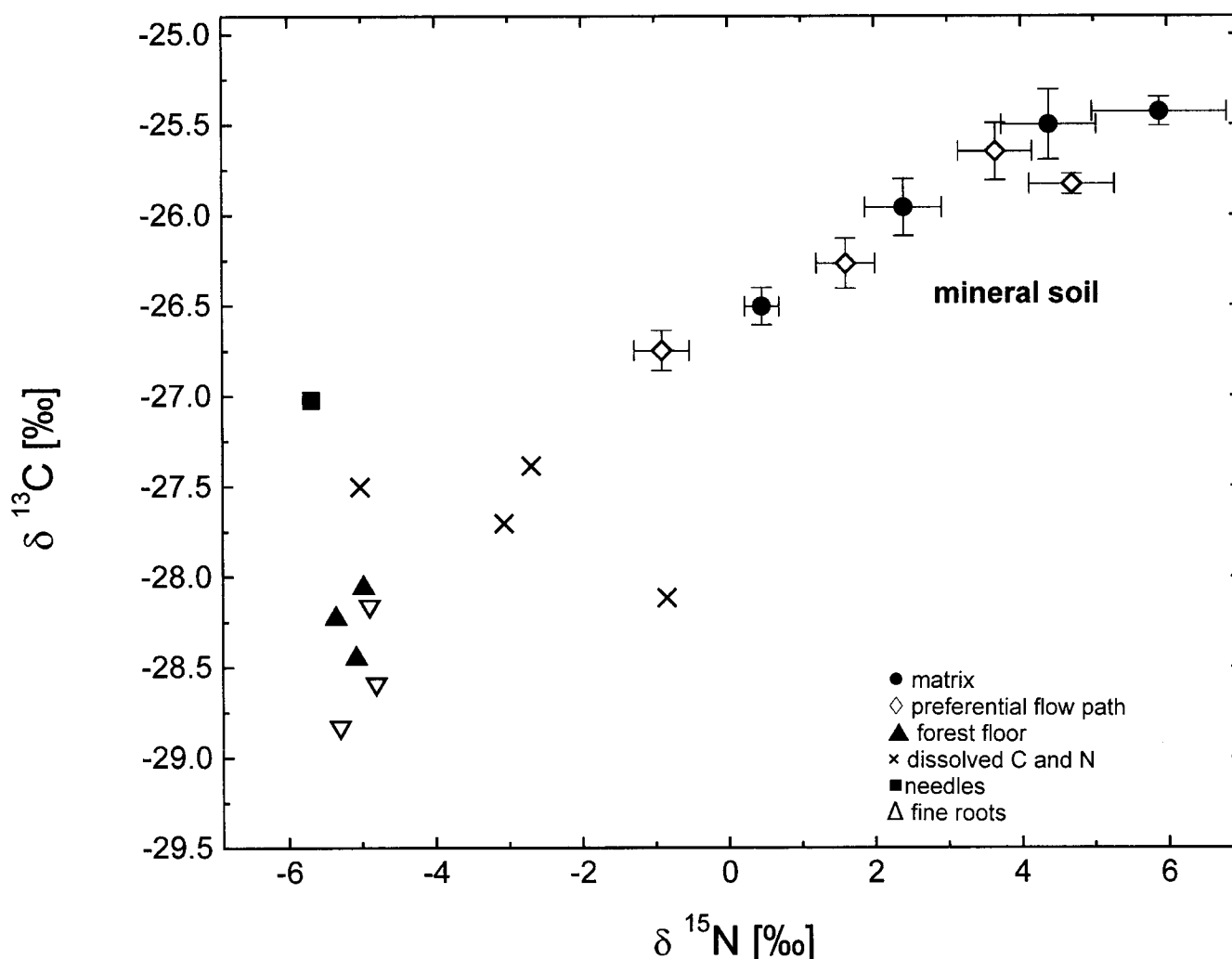


Fig. 7. Relationship between $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of leaves (solid square), organic layers (solid triangle), fine roots (open triangle), soil solution (cross), and soil material from preferential flow paths (open rhombus) and from soil matrix (solid dot). Shown are the mean and standard error of four plots. Samples of the mineral soil, soil solution, and fine roots were collected at different depths.

with the measured DOC inputs from forest floors into the mineral soil in other temperate forest ecosystems (Guggenberger and Zech, 1993; Currie, et al., 1996). As with the increased fine root production in preferential flow paths, this maximal possible increase of SOC through DOC illuviation is rather small compared with the observed total SOC accumulation along preferential flow paths.

The SOC accumulation in preferential flow paths of 740 to 960 g C m⁻² appears to be high even in comparison with the yearly C input through litter fall of about 200 g C m⁻² yr⁻¹ at this forest site (D. Hallenbarter, WSL, personal communication), which represents a typical litter input for temperate forests (Vogt et al., 1986). The remaining mass of forest litter after 3 yr of decay usually ranges between 15 and 30% (Melillo et al., 1989; Aber et al., 1990; Magill and Aber, 1998). Measured rates of SOC accumulation after forest establishment, with the maximal C input rate during soil development, range between 10 and 60 g C m⁻² yr⁻¹ (Schlesinger, 1990; Post and Kwon, 2000). Therefore, if all of the new

SOC is assumed to accumulate in the preferential flow paths, the C accumulation along preferential flow paths can, at most, account for 60 g C m⁻² yr⁻¹. Furthermore, it is unlikely that a stimulated mineralization of SOC in the soil matrix compared with that in the preferential flow paths contributed to the SOC enrichment in flow paths, since microbial biomass was higher along the preferential flow paths (Bundt et al., 2001).

The high SOC accumulation in preferential flow paths in comparison with potential SOC inputs strongly suggests that the increase of SOC in the predominant flow regions was a long-lasting process. Summing up over all potential contributors to SOC enrichment such as increased fine root biomass, enhanced DOC illuviation, and preferred incorporation of litter, indicates that SOC accumulation in the identified preferential flow paths lasted for at least 10 yr. This rough and conservative estimate is supported by radionuclide data from the same soil samples, which showed that the flow paths persisted for at least 40 yr (Bundt et al., 2000).

The high enrichment of SOC and the younger age of

SOC in preferential flow paths than in the rest of the soil suggest that preferential pathways play an important role in pedogenesis. The formation of new SOC in the mineral soil is probably not a vertical homogeneous process: at first, new SOC likely accumulates in soil regions that coincide with predominant flow pathways in the soil.

One implication of the SOC accumulation in the preferential flow paths would be a potentially higher sorption of contaminants with a high affinity to SOC, such as organic pollutants or heavy metals. This is consistent with other studies, which observed that (i) the sorption capacity of macropores is higher than the sorption capacity of the matrix, and that (ii) pesticide concentrations in the macropores decrease rapidly due to biodegradation (Stehouwer et al., 1994; Pivetz and Steenhuis, 1995; Mallawatantri et al., 1996).

Nitrogen Dynamics

The temporal variations of $\delta^{15}\text{N}$ after the ^{15}N labeling were higher in the preferential flow paths than in the matrix (Fig. 6). The tracer enrichment in the preferential flow paths was even higher than denoted by the $\delta^{15}\text{N}$ signature, because the total N concentrations were 25% higher in preferential flow paths than in the soil matrix. The low temporal resolution of the ^{15}N -tracer study can be responsible for the rather small preferential flow path effect. The N transport might have been faster than measured at the first sampling time 1 mo after the ^{15}N application. In the preferential flow paths, the $\delta^{15}\text{N}$ values decreased 1 mo after application, suggesting that the tracer front had already passed. In contrast, $\delta^{15}\text{N}$ values of the soil matrix still rose after 1 mo, which indicates a retarded arrival of the tracer front at locations in the soil matrix. The faster transport of NO_3 along preferential flow paths is consistent with the results of a field study with microsuction cups, which showed that the highest NO_3 concentrations in the soil solution of highly conductive zones were reached within hours after an NO_3 application and dropped back to the background concentrations within 3 d (Hagedorn et al., 1999). Nevertheless, the significantly higher increase of $\delta^{15}\text{N}$ in the preferential flow paths as compared with the matrix shows the importance of preferential flow paths for transport of nutrients and for plant nutrition. The latter is particularly important since the concentrations of plant roots in the preferential flow paths were also significantly higher than in the matrix (Table 1). As a result, the initial increase in $\delta^{15}\text{N}$ of the fine roots in preferential flow paths was more pronounced than in the soil matrix (Fig. 6).

After the initial increase of ^{15}N , $\delta^{15}\text{N}$ values decreased more in the preferential flow paths than in the soil matrix during the following 11 mo. This suggests an enhanced N cycling along preferential flow paths compared with the rest of the soil. However, we cannot distinguish whether this was because of a more pronounced leaching or root uptake of ^{15}N or to an increased illuviation of new unlabelled N.

CONCLUSIONS

Our results show that within a given soil horizon, SOC concentrations, total N concentrations, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values were not evenly distributed. Preferential flow paths, identified with a homogeneously applied dye tracer, had higher SOC concentrations compared with the rest of the soil. A ^{15}N -addition experiment led to a higher rise and a stronger decrease of the $\delta^{15}\text{N}$ values in the preferential flow paths than in the soil matrix. This suggests higher turnover rates in the preferential flow paths. Preferential flow paths were significantly depleted in ^{13}C and ^{15}N compared with the rest of the soil. This implicates that SOC from preferential flow paths was younger than SOC from the soil matrix. Potential pathways of SOC accumulation are an accelerated input of root-derived C and an increased illuviation of DOC. The SOC accumulation of 740 to 960 g C m⁻² in preferential flow paths compared with the soil matrix was much larger than the contribution from these potential sources. Therefore, accumulation of SOC in the identified preferential flow paths lasted for periods of more than 10 yr.

The findings of SOC enrichment along preferential flow paths have implications for the transport of reactive solutes through the soil. Sorption of organic pollutants and heavy metals strongly increases with increasing SOC concentrations. Thus, higher SOC concentrations in preferential flow paths may lead to an increased sorption and may therefore counterbalance the preferred transport of pollutants.

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